Embedded Systems – Question Bank (60 Questions × 5 Marks)

1 1. Define an embedded system and explain its key characteristics with examples. (5 Marks)
2 2. Differentiate between general-purpose computing systems and embedded systems. (5 Marks)
3 3. Explain the architecture of a typical embedded system with a neat block diagram. (5 Marks)
4 4. What are cyber-physical systems (CPS)? Explain their components and applications. (5 Marks)
5 5. Discuss the role of sensors and actuators in cyber-physical systems. (5 Marks)
6 6. Explain the interaction between computation, communication, and control in CPS. (5 Marks)
7 7. Define a real-time embedded control system. Give suitable examples. (5 Marks)
8 8. Differentiate between hard real-time and soft real-time systems. (5 Marks)
9 9. Explain the concept of deadlines and determinism in real-time systems. (5 Marks)
10 10. Describe the working of a real-time control loop with an example. (5 Marks)
11 11. Explain rate monotonic scheduling (RMS) with its assumptions. (5 Marks)
12 12. Explain earliest deadline first (EDF) scheduling and its advantages. (5 Marks)
13 13. Compare preemptive and non-preemptive scheduling in real-time systems. (5 Marks)
14 14. Define task, process, and thread in the context of real-time systems. (5 Marks)
15 15. Explain priority inversion and techniques to handle it. (5 Marks)
16 16. Explain I/O management in embedded systems. (5 Marks)
17 17. Differentiate between polling, interrupts, and DMA. (5 Marks)
18 18. Explain the role of device drivers in embedded systems. (5 Marks)
19 19. Discuss memory-mapped I/O and port-mapped I/O. (5 Marks)
20 20. What is an embedded operating system? List its key features. (5 Marks)
21 21. Compare embedded OS with general-purpose OS. (5 Marks)
22 22. Explain the services provided by an embedded operating system. (5 Marks)
23 23. Discuss popular embedded operating systems with examples. (5 Marks)
24 24. Explain inter-process communication (IPC) mechanisms in embedded OS. (5 Marks)
25 25. What is networking in embedded systems? Explain its importance. (5 Marks)
26 26. Explain TCP/IP stack with reference to embedded systems. (5 Marks)
27 27. Compare wired and wireless networking protocols used in embedded systems. (5 Marks)
28 28. Explain CAN protocol and its applications. (5 Marks)
29 29. Explain I2C protocol with timing diagram. (5 Marks)
30 30. Explain SPI protocol and its advantages. (5 Marks)
31 31. What is a System on Chip (SoC)? Explain its architecture. (5 Marks)
32 32. Differentiate between microcontroller and SoC. (5 Marks)
33 33. Explain the role of IP cores in SoC design. (5 Marks)
34 34. Discuss advantages and challenges of SoC-based designs. (5 Marks)
35 35. Explain memory subsystem in embedded systems. (5 Marks)
36 36. Differentiate between ROM, RAM, Flash, and EEPROM. (5 Marks)
37 37. Explain cache memory and its importance in embedded systems. (5 Marks)
38 38. Discuss memory hierarchy with a neat diagram. (5 Marks)
39 39. Explain bus structure in embedded systems. (5 Marks)
40 40. Differentiate between data bus, address bus, and control bus. (5 Marks)
41 41. Explain AMBA bus architecture. (5 Marks)
42 42. Discuss the role of bus arbitration techniques. (5 Marks)
43 43. Explain interfacing protocols used in embedded systems. (5 Marks)
44 44. Discuss UART interfacing with microcontrollers. (5 Marks)
45 45. Explain peripheral interfacing with suitable examples. (5 Marks)
46 46. What is power management in embedded systems? (5 Marks)
47 47. Explain techniques used for power optimization. (5 Marks)
48 48. Discuss dynamic voltage and frequency scaling (DVFS). (5 Marks)
49 49. Explain embedded system software development life cycle. (5 Marks)

50 50. Discuss code optimization techniques for embedded software. (5 Marks)
51 51. Explain the importance of compiler optimization in embedded systems. (5 Marks)
52 52. What is concurrent programming? Explain its need in embedded systems. (5 Marks)
53 53. Explain race condition and deadlock with examples. (5 Marks)
54 54. Discuss synchronization mechanisms used in concurrent programming. (5 Marks)
55 55. Explain real-time constraints in concurrent embedded systems. (5 Marks)